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ABT Site 6.1

IPv6

Engineering

A6V14633054_en--_a_reduced
2025-09-23

Smart Infrastructure

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Edition: 2025-09-23

Document ID: A6V14633054_en--_a_reduced

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1 Cyber security disclaimer

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https://www.siemens.com/cert/ => 'Siemens Security Advisories'.

2 Introduction

This document describes IPv6 in a building automation system to varying degrees of detail:

- Introduction to IPv6
- IPv4 versus IPv6
- IPSec
- Obstacles to adopting IPv6
- Cybersecurity

Target audience

This document is intended for internal engineers, IT specialists, and commissioning engineers. Due to the sensitive nature of the topic (cybersecurity), it is not intended for publication to a general audience. We welcome feedback from the field.

Much of this document covers information or functionality not released to the general public but that engineers may require to work with the system. We plan to transfer these sections to the relevant documents as they are released.

3 IPv6

The main motivation for introducing IPv6 is that the available IP addresses under the 32-bit IPv4 protocol are running out (or already have). There are only 4.3 billion unique IPv4 addresses.

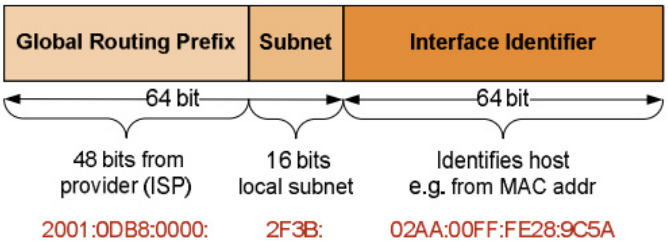
IPv6 offers more than just nearly unlimited addressing. It is scalable and easier to manage since it does not require complex workarounds, such as Network Address Translation. It is also future proof, supporting mobile networks and cloud computing.

Other improvements include:

- Simplified headers (40 bytes and fewer fields)
- No NAT required
- Autoconfiguration (both stateless (SLAAC) and stateful (DHCPv6))
- IPsec support*
- Improved handling of multicast and neighbor discovery
- Multicast and Anycast reduce network congestion
- IPv6 routers do not fragment packages, improving performance

** IPsec functionality in the IPv6 stack is mandatory, but its use is not. IPsec is also not implemented automatically; it must be configured.*

3.1 Address syntax



IPv6 uses a 128-bit address written as eight groups of 4 hexadecimal digits. The address is divided into 2 parts:

- Network prefix: Assigned by an ISP (48 bits + 16 bits for subnetting) or regional internet registry
- Interface Identifier (IID): The remaining 64 bits that identify a device within a subnet.

Internet service providers or regional internet authorities generally allocate a /48 prefix.

3.2 Address types

There are three categories of IPv6 addresses:

Unicast addresses send packets to a specific device and are categorized into the following subtypes:

- Global Unicast Addresses (**GUAs**) are routable on the public internet and used for global communication.
- Link-Local addresses (**LLAs**) are automatically configured for communication within a single link (i.e., a single layer-2 domain such as an Ethernet segment).
- Unique local addresses are intended for internal communication on private networks and are not routed on the public internet.

Multicast addresses identify a group of addresses. Packets sent to a multicast address are delivered to all interfaces in the group. They are used for streaming or service discovery.

Anycast addresses are assigned to multiple interfaces. However, packets are routed to the nearest address (based on network topology) and are used for load balancing or redundancy.

Type	Range	Start
All/default route	::/0*	::
Unspecified	::/128	::
Host (loopback)	::1/238	::1
GUA	2000::/3	2000::
Unique local	fc00::/7	fe80::
LLA	fe80::/64	Fe80::
Multicast	ff00::/8	

* The backslash indicates the number to significant bytes (from left to right).

3.3 Abbreviating addresses

There are two rules for abbreviating IPv6 addresses:

1. Omit leading zeros.
2. Use a double colon (::) to represent quartets containing all zeros in sequence (this can only be used once per address).

For example, an LLA of fe80:cafe:**0000:0000:0000:00**ae:**0**ffe:**000**1 can be abbreviated to: fe80:cafe::ae:ffe:1. The highlighted hex numbers are abbreviated.

3.4 Network prefix

The prefix length designates the number of bits used for the network components (global routing prefix plus subnet ID); the remaining bits are available for host addressing. Prefix /64 is standard for LANs and matches the length required by Stateless Address Autoconfiguration (SLAAC) to assign IP addresses.

3.5 Subnetting process

1. Write down the allocated prefix: For example, 2001:DB8:CAFE::/48 (64-16 bits used for the actual subnet)
2. Select subnet size: As a rule /64 (SLAAC). SLAAC uses the lower 64 bits for the Interface Identifier, so that IPv6 requires /64 subnets as a strong rule.
3. Allocate subnets: The bits between the allocated prefix (16 bits) are used for the subnet IDs (one-up). For example, 2001:DB8:CAFE::/64 is the first; 2001:DB8:CAFE:0001::/64 is the second, etc.
4. The remaining 64 bits are for IIDs.

In our example, the Internet Service Provider assigned 48 bits for his global routing prefix. This leaves 16 bits for subnets. Each of these 65,537 (2^{16}) possible subnets is a /64 network:

- 2001:DB8:CAFE:0000::/64 (first subnet)
- 2001:DB8:CAFE:0001::/64 (second subnet)
- • … up to 2001:CAFE:1234:FFFF::/64 (65536th subnet)

Although tools are available for calculating subnets, /64 alone allows for a total of 65,536 (2^{16}) subnets.

3.6 Best practices for subnets

1. Use /64 subnets (SLAAC compatibility).
2. Reserve portions of the subnet ID space for future use.
3. Document.
4. Do not overcomplicate assignments (one-up).

3.7 EUI-64

While SLAAC is the mechanism to automatically generate a unique 64-bit Interface Identifier, there are several algorithms that can implement it. EUI-64 is one such basic method: EUI-64 (Extended Unique Identifier 64) automatically generates a 64-bit IID for SLAAC. It is combined with the 64-bit network prefix to form a 128-bit address (e.g., 2001:DB8:CAFE:0001::/64 + IID). EUI-64 uses the 48-bit MAC address to create the IID and self-configures IPv6 addresses without the need for a separate DHCPv6 server.

Syntax

1. Start with the 48-bit MAC address (six pairs of hex digits, e.g., 5E:4D:3C:2B:1A:00). The first 24 bits identify the manufacturer, and the remaining 24 bits are unique to the device.
2. Insert FF:FE. Split the MAC address into two parts and insert FF:FE in the middle. For example, 5E:4D:3C:2B:1A:00, this becomes 5E:4D:3C:**FF:FE**:2B:1A:00.
3. Flip the seventh bit (universal/local bit). In our example, hexadecimal 5E converts to binary as 0101 1110 and now becomes 0101 1100 or 5C.

   The final address is: 5C:4D:3C:FF:FE:2B:1A:00.
4. Append the IID address to the network prefix of (with subnet 0001) of 2001:CAFE:1234:0001:**5C4D:3CFF:FE2B1A:00**.

This address is unique to the device and SLAAC-compatible and does not require DHCPv6. It also works well in IPv6 /64. The address is, however, fixed to the MAC address and can be tracked across networks, which can be regarded as a privacy issue. Some devices or networks disable EUI-64 in favor of random IIDs or DHCPv6 (and manual configuration). EUI-64 is being replaced by privacy extensions in modern IT networks to provide random IIDs.

3.8 Privacy extensions

Privacy extensions use a cryptographic algorithm or random number generator to generate random IIDs. The IID is added to the network prefix, resulting in an address 2001:CAFE:1234:0001:8765:4321:FEDC:B9A7 when used with the above network prefix and IID of 8765:4321:FEDC:B9A7.

3.9 Multicast addresses in BACnet/IPv6

In contrast to IPv4, IPv6 does not use broadcasts. Instead, IPv6 uses what are referred to as multicast addresses. Conceptionally, the multicast addresses represent a group of devices that can be reached by an assigned multicast address. Unlike IPv4 devices, an IPv6 device typically has more than one IP-address. The IPv6 standard views multicast addresses as something that a device can join and "leave", in other words, it is a dynamic process during runtime. As a rule, multicast addresses are set as static addresses when configuring a device.

Consult with IT-departments on whether multicast communication is supported (or not) in a project and not the type. The Neighbor Discovery Protocol (NDP) in IPv6 heavily relies on multicast and admins usually allow link-local multicast (ff02::/16) but often block or filter other multicast scopes.

The BACnet standard has pre-reserved some common multicast addresses for BACnet communication:

FF02::BAC0	Link-local scope multicast address: Communicates within the local network segment and is not routable.
FF04::BAC0	Admin-local; The smallest scope that is configured administratively.
FF05::BAC0	Site-local scope multicast address: For communication within a specific site or administrative domain.

FF08::BAC0	Organization-local scope multicast address; This is the largest scope for communication within an organization.

4 IPv4 versus IPv6

IPv6 offers several improvements over IPv4 including a larger address range, enhanced performance, and built-in security features.

4.1 Comparison of IPv4 vs. IPv6

Function	IPv4	IPv6
Total number of addresses	2^{32}	2^{128}
Usual format	Decimal	**Hexadecimal**
Header	20-60 bytes (variable)	40 bytes (fixed)
NAT	Required	Not required
Security	IPsec (optional)	IPsec*
Privacy extensions	No	Yes
Fragmentation	By routers and endpoints	Endpoint only
Autoconfiguration	DHCP	SLAAC

* IPsec functionality in the IPv6 stack is mandatory, but using IPsec is not. In other words, IPsec is not automatically implemented; it must be configured.

4.2 Structure and configuration

IPv6 uses hexadecimal format (the same as MAC addresses). In other words, MAC addresses can be used in IPv6 addresses without translation and are easier to work with. IPv6 uses stateless address autoconfiguration (SLAAC) and larger subnet allocations to simplify address management.

4.3 Performance

IPv4: Variable header sizes increase processing, and NAT introduces latency and complexity to routing.

4.4 Security

The limited address space and reliance on NAT make IPv4 vulnerable to spoofing and other attacks. IPv6 supports IPsec* and the larger address field makes scanning attacks more difficult. The device MAC address is often fixed in IPv6 and therefore easier to spoof (if not encrypted or randomized). IPv6 reduces the need for Network Address Translation (NAT) which generally complicates security policies and end-to-end visibility; security policies become more transparent using globally unique addresses.

*This does not imply that IPsec is always used, but it ensures that all IPv6-capable devices can support secure end-to-end encryption and authentication.

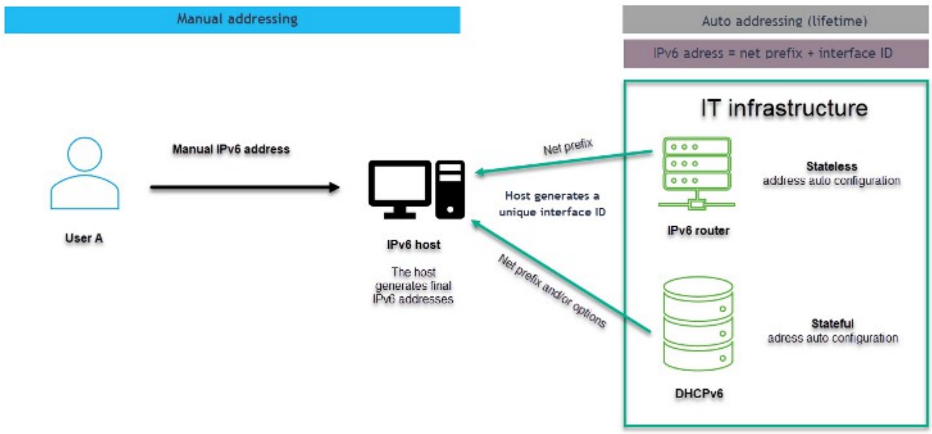
4.5 Fragmentation

In IPv4, IP-routers along the communication path can fragment packets if a packet is larger than the next link's Maximum Transmission Unit (MTU). The destination endpoint then reassembles such fragmented packets back into their original format. In IPv6, packets can also be fragmented, but only by the sending endpoint: IP-routers along the path do not perform fragmentation; instead, they notify the sender if a packet is too large (ICMPv6 "Packet Too Big" message).

5 Address allocation

There are two methods for allocating addresses automatically in IPv6:

- Stateless address autoconfiguration (SLAAC)
- Dynamic host configuration protocol (DHCPv6)



5.1 SLAAC

SLAAC automatically assigns IPv6 addresses using router advertisements and is stateless, no server tracks the assignments. SLAAC provides an IPv6 address and Default Gateway, and devices self-configure by combining a network prefix (from the RAs) with an interface identifier (using the device's MAC address). However, the MAC address makes addresses predictable, and thus susceptible to attacks (privacy extensions resolve this issue). Devices generate their own IPv6 addresses automatically with SLAAC by:

- Using the network prefix from Router Advertisements (RA).
- Generating an Interface Identifier (IID), often via EUI-64 or randomization.

SLAAC is best suited for small-to-medium networks with little to no service infrastructure or configuration since it does not require a DHCPv6 server.

5.2 DHCPv6

DHCPv6 is a stateful protocol that assigns IPv6 addresses and other network information from a centralized server which tracks address assignments. DHCPv6 provides most configuration options, such as IPv6 addresses, DNS servers, and domain names. It requires a separate DHCPv6 server to manage and distribute addresses and configurations. Addresses are controlled by the server and administrators can assign specific addresses, reserve addresses, or manage address pools. DHCPv6 can also randomly assign addresses or supply temporary ones. DHCPv6 is ideal for large or managed networks that require centralized control and configuration. It is the preferred solution for enterprise networks requiring strict control of address assignments, detailed configuration, or integration with management systems (e.g., Desigo CC).

Each option can operate separately, coexist, or be combined into a hybrid model on the network (see Cybersecurity [→ 21]).

SLAAC and DHCPv6 are not mutually exclusive. In fact, they can complement each other. For this, DHCPv6 can be used in two modes:

- Stateful DHCPv6: The server assigns the IPv6 address and other options (like DNS).
- Stateless DHCPv6: The server only provides additional information (such as DNS servers), while the client uses SLAAC for addressing

In this way, a device can use SLAAC to generate a unique IPv6 address and then use stateless DHCPv6 to find a DNS server.

It is also possible to first use SLAAC but just for bootstrapping purposes and then use a stateful DHCPv6 to re-assign a new IPv6 address to the device. In this way, the IPv6 address becomes stateful, as it was assigned by the DHCPv6 server which tracks it in its database. However, this combination is less common than the SLAAC + stateless DHCPv6.

5.3 Static addresses

Static addresses are generally used whenever the operator needs to control a stable and predictable network. While dynamic addresses are usually preferable, there are situations where static addresses are more appropriate, especially given the large number of available addresses:

- Servers and services: Devices hosting critical services
- Network infrastructure
- Firewalls and access control
- IoT or embedded devices
- Testing or development
- Legacy or specific applications

Dynamic IPv6 addresses (be it SLAAC or DHCPv6) can change over time, thus complicating these tasks.

If SLAAC is used without intentional randomization, i.e., using the basic EUI-64 mechanism, the generated addresses become deterministic or static over time.

6 Migration

IPv6 is not backwards compatible with IPv4: IPv6 devices cannot communicate directly with IPv4-only devices without the use of transition mechanisms.

There are three basic methods for transitioning IPv4 to IPv6: NATing (e.g., by 464XLAT or NAT64), NPTv6, and dual-stack. Dual-stack is the recommended transition strategy for customers that want to transition from a primarily IPv4 network to a primarily IPv6 network: Start by adding IPv6 to the hosts and then gradually disabling IPv4.

Dual-stack means that each device can speak both IPv4 and IPv6 in parallel. Instead of routing a message on the IP-layer (not possible since both variants are incompatible), it is the application layer on the device that binds both network variants and reacts accordingly. Typical example: A webserver can bind both an IPv4 and an IPv6 network in parallel and thus handle incoming web access from both network types.

NATing means that some mechanisms are applied to rewrite packets from IPv4 to IPv6 and, vice versa, to emulate the non-existing IPv4-to-IPv6 routing mechanism over the IP layer. Such NAT-mechanisms are naturally limited in what they can achieve and require all devices in a network to support the same NAT-mechanism.

6.1 464XLAT

464XLAT is a client-server solution that enables IPv6-only clients to access IPv4-only services. It combines two components:

- CLAT (Customer-side Translator): Translates IPv4 packets to IPv6 on the client side.
- PLAT (Provider-side Translator): Translates IPv6 packets back to IPv4 on the server side.

464XLAT is widely adopted in mobile networks to provide IPv6-only connectivity to its users.

6.2 NAT64

NAT64 translates between IPv6 and IPv4 and vice versa, enabling IPv6-only devices to communicate with IPv4-only devices.

Use case:

- IPv6-only clients accessing IPv4-only servers and vice versa.
- Stateful NAT64 scales for multiple clients via port translation.

However, the method breaks the end-to-end principle in stateful mode, is highly complex and requires DNS64 integration which adds complexity. Additionally, stateful NAT64 increases resource use. It is not used for IPv6-to-IPv6 communications. Nonetheless, this may go hand-in-hand with DNS64, which allows an IPv6-only device to access an IPv4-only name server. Desigo devices do not support NAT64.

6.3 NPTv6

NPTv6 (Network Prefix Translation for IPv6) is not a NAT-like IPv6-to-IPv4 translation mechanism, but an IPv6-to-IPv6 translation for prefixes, typically used in multi-homing or renumbering scenarios. It supports the use-case of an IPv6<->IPv6 migration scenario, i.e., if a site changes from one ISP to another ISP. In that case, NPTv6 allows an automatic exchange of the global routing prefix without the need for reconfiguring all equipment IPv6 addresses on the site.

NPTv6 statically maps one IPv6 prefix to another while preserving the host portion, facilitating communication across networks with different prefixes. It is stateless and does not translate ports and is used to translate IPv6 prefixes for multi-homing or network isolation. The stateless protocol is highly scalable and preserves the

IPv6 end-to-end principle. It is simple and requires minimal overhead due to its stateless operation and is ideal for IPv6-only networks that require different prefixes. Desigo devices do not support NPTv6.

6.4 Dual stack

This method is used in the Desigo System to transition to IPv6. Both protocols operate simultaneously on the same devices, providing full support to both IPv4 and IPv6 without affecting compatibility. Devices use the necessary protocol and no translation is involved. This method is best suited for mixed environments that support both protocols during the transition. It requires both A (IPv4) and AAAA (IPv6) records for services that rely on DNS resolution. The two protocols increase resource usage, so operators must take care to regularly update both IP protocols to the latest versions. Each protocol must be secured separately.

6.5 BACnet/SC

For customers who consider BACnet/SC important for IPv6, we recommend a phased approached to BACnet/SC.

There are two possible migration approaches that span across multiple years / market releases:

Phased from BACnet/IPv4 to BACnet/IPv6 before migrating to BACnet/SC @ IPv6 (preferred)

In a phased approach, the system starts with BACnet/IPv4, phases in BACnet/IPv6 before implementing BACnet/SC IPv6. This is the preferred method.

Direct from BACnet/IPv4 to BACnet/SC IPv4 and (direct) to BACnet/SC@ IPv6

In general, the transition to IPv6 typically begins in the backbone of an organization's network infrastructure—such as core routers and firewalls - and gradually extends through IT systems to the automation level, including building automation controllers and operational technology (OT). It makes sense to ensure an operational IPv6 system over the network, before attempting to make the jump from BACnet/SC IPv4 to IPv6.

NOTICE	
i	**BACnet/SC does not support IPv6 on Desigo devices.** We recommend a phased approach to IPv6 over BACnet/SC (see above). There is no timeline at this time for implementing IPv6 over BACnet/SC.

7 IPsec

IPsec (Internet Protocol Security) is a protocol that provides authentication, data integrity, and encryption to secure communications over IP-layer 3. IPsec is optional in IPv4 and supported in IPv6. IPsec functionality in the IPv6 stack is mandatory but its use is not. IPsec is not automatically implemented.

While IPsec is quite common on Virtual Private Networks (VPNs), it is rarely used for device communication today: TLS (on layer 4) and application level protocols (L7) on top (such as https or TLS secured WebSockets in the case of BACnet/SC) have largely replaced IPsec in the field. The following are the basic requirements for implementing IPsec over IPv6 for information purposes only:

7.1 Prerequisites

Network	IPv6 enabled
Hardware/software	Supports IPv6
Key management	• Manual (simpler, but less secure) • Automated (via Internet Key Exchange (IKEv2)
Certificate or PSKs	• Certificates for scalability and security
Firewall rules	IPv6 firewall rules allow IPsec traffic

The preferred options are highlighted below.

7.2 Authentication

Authentication header (AH)	For data integrity and authentication, but not encryption. It is rarely used and has been mostly replaced by ESP.
Encapsulating security payload (ESP)	ESP is an extension header that ensures confidentiality, integrity, and authentication.

7.3 Modes

ESP has 2 modes: Transport and tunnel mode

Transport mode	Only the payload is encrypted. The IPv6 header remains unchanged. For host-to-host communication.
Tunnel mode	Encrypts the original IPv6 packet and encapsulates it in a new IPv6 header. This method is primarily used in VPNs for gateway-to-gateway communication.

7.4 IKE2

IKE2 generates and manages security associations to protect network traffic. These include encryption algorithms and authentication methods.

7.5 Encryption methods

AES-CBC	Block cipher mode; requires a separate integrity algorithm.
AES-GCM (preferred)	Authenticated encryption that combines encryption and integrity, reducing overhead.
AES-CCM	Used in resource-constrained devices.
ChaCha20-Poly1305	High-performance, secure stream cipher: Well suited for modern or lightweight devices, such as those using IoT over IPv6 networks.

AES-GCM is the preferred method for modern IPv6 networks.

Authenticated encryption with associated data (AEAD) combines encryption and data integrity into a single operation, making it well-suited for use in IPsec ESP.

7.6 Verification

HMAC-SHA-256	Secure hash algorithm (SHA) has a 256-bit output. It is often paired with AES-CBC for encryption and is widely supported.
HMAC-SHA-384	The 384-bit output provides a higher level of security but requires more processing. This option is less common, but it is used in high-security IPv6 deployments.
HMAC-SHA-512	It offers the highest security among the HMAC algorithms and is suitable for critical, resource-intensive IPv6 applications.
HMAC-SH1	(Legacy) No longer used (security vulnerabilities).
HMAC-MD5	(Legacy). Do not use.

8 Obstacles to adopting IPv6

IPv6 has been around for over 25 years and yet only about 40% of traffic on the Internet is routed via IPv6 (https://www.google.com/intl/en/ipv6/statistics.html). This is due to a number of obstacles that companies must overcome to implement the protocol, including:

1. Existing infrastructure and compatibility
2. Costs
3. No obvious business incentive
4. Complexity and risk of misconfiguration
5. Limited IPv6 content and support
6. Transition overhead

8.1 Existing structure

The greatest barrier to adopting IPv6 is that IPv4 and IPv6 are not compatible and, since an IP-router IPv4 <-> IPv6 does not exist, the success of IPv4 and its installed base now prevent a smooth transition. Most existing devices, software, and network infrastructure are designed for IPv4 and do not fully support IPv6 which forces companies to undertake costly upgrades.

Mitigation strategies include: Employ dual-stack networks (Desigo does not support tunneling to bridge IPv4 and IPv6 networks) and prioritize IPv6 procurement.

Migration is currently in phases: IPv4 → Hybrid IPv4 + IPv6 → IPv6-only

We recommend starting with dual-stack deployment to allow both protocols to operate concurrently. This may not always be practical down to the device level as this would require implementing it on every device on the IP-network(s) and, at the same time, binding both IP-variants on the devices. To avoid this, some application-layer routers or gateways (L7) can form a perimeter-based transition between the two network protocols.

A BACnet router that supports BACnet/IP and BACnet/IPv6 is an example: It has a built-in dual stack that binds to an IPv4 network and an IPv6 network and can translate between BACnet/IP on the IPv4-side of the stack and BACnet/IPv6 on the IPv6-side. It serves as a gateway between a BACnet/IP network of devices and a BACnet/IPv6 network of devices and neither side is required to support its own dual stack.

Of course, the end goal is to engineer and operate IPv6-only environments.

To this end, it is crucial to ensure that the system as a whole is compatible and performance is maintained across IPv6 or hybrid IPv4+IPv6 environments. Finally, operating IPv4 and IPv6 concurrently increases testing and maintenance overhead.

8.2 Costs

The transition to IPv6 requires significant human resources and money.

The major cost drivers are staff training and configuration complexity that requires specialized knowledge.

Mitigation: Use open-source tools to reduce software costs and implement pilot projects to spread the cost out over time and create gateways (see above), such as the BACnet router to support the hybrid solution until the devices need to be replaced (due to failure or outdated technologies).

We recommend spreading the cost over time by starting with pilot deployments and a hybrid IPv4+IPv6 approach to avoid immediate full replacement.

8.3 No obvious business incentives

Companies do not see an immediate need to transition because Network Address Translation (NAT) adequately meets current needs.

Mitigation: Eliminating NAT results in cost savings over the long term and reduces complexity.

However, the GSA order (U.S. General Services Administration), aligned with the White House OMB Memorandum M-21-07 highlights key incentives related to IPv6:

- IPv6-only infrastructure for all new federal contracts as of July 1, 2023. In other words, no IPv4-only devices are permitted in new topologies as of this date.
- Any defects or replacements to existing (deployed) hardware must prioritize IPv6-capable replacements, where possible.

In a nutshell, this order enforces a move to IPv6 technology across all federal agencies and any IP-enabled assets in their networks. No explicit distinction is made between IT and OT technology, so that this also applies to IP-enabled and embedded devices on OT (operational technology) systems as used in Building Automation Control Systems (BACS), i.e., to move from IPv4 to IPv6.

Overall cost and cybersecurity are the primary drivers for the move: *standards bodies and leading technology companies began migrating toward IPv6-only deployments, thereby eliminating complexity, operational cost, and threat vectors associated with operating two network protocols.*

The order also provides a timeline for migrating to IPv6 but with the understanding that there will be systems that for whatever reason are not be able to comply with the roadmap:

- No later than fiscal year (FY) 2023, all new *networked federal information systems are IPv6-enabled at the time of deployment*
- *At least 20% of IP-enabled assets on federal networks are operating in IPv6-only environments by the end of FY 2023*
- *At least 50% of IP-enabled assets on federal networks are operating in IPv6-only environments by the end of FY 2024*
- *At least 80% of IP-enabled assets on federal networks are operating in IPv6-only environments by the end of FY 2025*
- *Identify and justify federal information systems that cannot be converted to use IPv6 and provide a schedule for replacing or retiring these systems*

It seems that this high-level executive order and its timeline focus on IT (Information Technology) devices, which typically have a shorter expected life span of ~2 to 5 years, versus OT (Operational Technology) devices with typical life spans in the range of ~20+ years. Still, the IPv6-only argument that new sites may not even have an IPv4/IPv6 dual-stack anymore, is understandable and needs to be considered.

The executive order defines some critical distinctions:

- **IPv6-capable** refers to a system or service that has correctly implemented a complete set of IPv6 capabilities but that are not necessarily enabled.
- **IPv6-enabled** refers to a system or service in which the use of IPv6 is turned on for production use.
- **Native IPv6** refers to direct support of IPv6 in a system or service without requiring the use of IPv4 for basic communication.
- **IPv6-only** refers to the state of an operational system or service when IPv4 protocol functions (addressing, packet forwarding) are not in use.

Further clarification:

- **Dual stack** refers to devices operating IPv4 that cannot work with IPv6 but their firmware can be upgraded to operate IPv6.
- **IPv6-capable** means that a device has also implemented functionality for IPv6 but is not necessarily using it. A typical use case would be an existing building that is operating IPv4, but the customer wants to migrate to IPv6 at some point without the need to physically replace or upgrade devices: The devices would only require reconfiguration.
- **IPv6-enabled** means that a device actually uses IPv6 for runtime operations. However, during the initial engineering and commissioning phase or maintenance phases, it may still require the use of IPv4 (in other words, is a dual-stack device). This is the Desigo approach to migrating [→ 14] from IPv4 to IPv6.

- **Native IPv6** means a device primarily communicates with IPv6 but may use IPv4 sporadically for some minor functions (in other words, it is also a dual-stack device)
- **IPv6-only** means a device cannot rely on IPv4 as the underlying LAN infrastructure does not support or actively blocks IPv6.

8.4 Technical complexity

New features such as SLAAC, privacy extensions, and multicast have a steep learning curve. Misconfigured IPv6 networks can expose devices to attacks and dual stacks increase complexity.

Mitigation: Implement training and certification programs and use automated configuration programs to reduce human error. Also, implement strict IPv6 firewall rules to mirror IPv4 security.

8.5 Translation overhead

Dual-stack, tunneling, and translation are complex processes and can impact performance.

Mitigation: Use dual-stack based application layer routers / gateways (e.g., BACnet/IP <-> BACnet/IPv6 routers) to support a phased migration and peaceful co-existence of both IPv4- and IPv6-based L7 protocols. Prioritize IPv6 over IPv4 transition mechanisms and optimize tunneling. Phase out IPv4 controlled environments.

9 Cybersecurity

From a cybersecurity viewpoint: SLAAC alone is the least secure method but it is also the easiest to configure. Stateful DHCPv6 is the most secure method. SLAAC + stateless DHCPv6 is secure but is difficult to track in the event of cyberattacks or in environments requiring regular audits.

BACnet/SC does not currently support IPv6.

	SLAAC	**Stateful DHCPv6**	**DCHPv6 stateless + SLAAC**
BACnet/SC	Not supported	Not supported	Not supported
Configuration	Auto configuration	Requires considerable engineering	Minimum engineering
Single point of failure	No	No	Yes
Device support	IPv6	Not all devices	Not all devices

SLAAC	Best suited for smaller networks
Stateful DHCPv6	Best suited for enterprise networks that track and audit network traffic
SLAAC + stateless DHCPv6	Best suited for large networks with lots of devices (where tracking is not a priority). Desigo PXC5/7 devices support both stateful and stateless DHCPv6 operation as of ABT Site 6.1.

9.1 SLAAC

9.1.1 Vulnerabilities

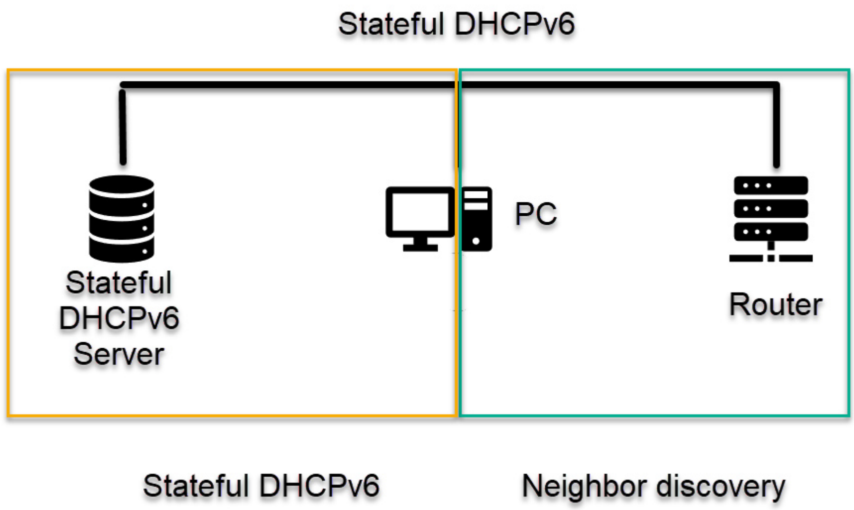
1. Authentication: Weak
2. Router advertisements (RAs) are not authenticated by default and therefore vulnerable to RA attacks such as sending fake RAs to redirect traffic (i.e., man-in-the-middle attacks).
3. Generates interface IDs based on MAC address. The address is static (i.e., predictable) and permits tracking of devices across networks.
4. Stateless protocol with no central record of addresses, rendering it difficult to detect unauthorized devices or track address use.
5. RAs provide a large attack surface: RA flooding or prefix spoofing can disrupt the network or even redirect traffic.

9.1.2 Mitigation

1. Secure Neighbor Discovery (SEND) authenticates RAs using cryptographic methods (complex and rarely used).
2. Devices can use privacy extensions (generally enabled by default, but check) to randomize temporary interface identifiers.
3. Network monitoring tools such as RA Guard block unauthorized RAs (requires additional engineering).
4. RA Guard and SEND reduce attack surfaces (not universally deployed).
5. RA Guard and SEND prevent RA flooding (not always implemented).

9.2 DHCPv6

Stateful DHCP assigns and tracks addresses and is similar in concept to DHCPv4. The process is 2-staged: 1) Neighbor discovery, followed by 2) Stateful DHCPv6.



Step	Route	Stateful	Neighbor discovery
1	PC* to router		Router solicitation
2	Router to PC		Router Advertisement Message 1. M-flag = 1 2. 1 2. A-flag = 0 to enable
3			Default gateway
4	PC to server	DHCPv6 SOLICIT	
5	Server to PC	DHCPv6 ADVERTISE	
6	PC to server	DHCPv6 REQUEST	
7	Server to PC	DHCPv6 REPLY	

* PC is any device on the network requiring addressing.

9.2.1 Vulnerabilities

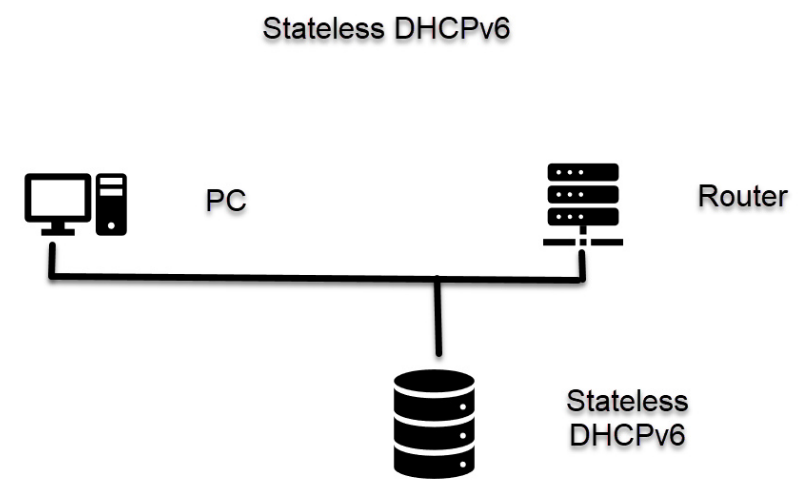
1. Authentication: Strong.
2. DHCPv6 server attacks can assign incorrect addresses, potentially enabling traffic interception or DoS attacks.
3. Servers assign addresses from a pool and may be predictable or tied to a device ID if not properly configured.
4. Centralized servers become a single point of failure and potential target (DoS attacks).
5. DoS attacks against the DHCPv6 server can prevent devices from obtaining addresses.

9.2.2 Mitigation

1. Authentication mechanisms verify server and client (not always enabled by default).
2. DHCPv6 can be configured to assign temporary or randomized addresses and anonymous DHCPv6 to limit client identifier exposure.
3. DHCPv6 snooping or similar prevents rogue servers. Only trusted servers are permitted to assign addresses.
4. DHCPv6 snooping, authentication, and network segmentation reduce the attack surface (single server has a smaller surface as compared to distributed RAs).
5. Rate-limiting, redundant servers, and DHCPv6 snooping mitigate DoS attacks.

9.3 SLAAC + stateless DHCPv6

SLAAC + DHCPv6 is a hybrid approach that applies the strengths of both protocols while addressing specific limitations.



Step	Route	
1	PC to router	Router solicitation (RS)
2	Router to PC and server	Router Advertisement (RA) 1. A-flag = 1 2. O-flag = 1 3. M-flag = 0
3	Stateless DHCPv6	Stateless auto-configuration (SLAAC)
4	PC to network	Duplicated Address Detection (DAD)
5	PC to DHCPv6 server	DHCPV6 SOLICIT
6	DHCPv6 to PC	DHCPV6 ADVERTISE
7	PC to DHCPv6 server	DHCPv6 INFORMATION-REQUEST
8	DHCPv6 server to PC	DHCPv6 REPLAY

9.3.1 Benefits

1. Authentication: Strong, but limited ability to track traffic
2. Simplicity: SLAAC does not require a server to manage addresses, reducing complexity while stateless DHCPv6 requires only minimum server resources (since it does not track address leases).
3. Comprehensive: Devices have the requisite parameters without stateful DHCPv6.
4. Scalable: Can be used on small and larger systems that require centralized DNS but where address management is non-critical. Works well on networks with various devices since most IPv6-capable devices support both protocols (check with network administrator).
5. Reduced dependency: Stateless DHCPv6 does not maintain a lease database. This reduces the risk of a server failure impacting addresses.
6. Privacy: SLAAC uses privacy extensions for randomized temporary address; DHCPv6 supports anonymous modes to minimize ID exposure.

SLAAC

Autonomously configures IPv6 addresses.

1. Receives RAs from the network prefix and default gateway from the router.
2. Generates interface ID to form a complete IPv6 address.

Stateless DHCPv6

Devices query DHCPv6 server for configuration parameters, including:

1. DNS server addresses
2. Domain search lists
3. Optional settings (e.g., NTP servers)

9.3.2 Limitations

1. Address control is limited.

   Administrators cannot enforce specific addresses or track assignment.
2. Security

   The combination is only as strong as the underlying mitigation. DHCPv6 servers remain at risk without appropriate security.
3. Configuration: More complex than just SLAAC.
4. MAC addresses are fixed and therefore easy to spoof.

9.4 Cybersecurity best practices

Enable IPv6-Aware firewalls

- Deploy IPv6 firewalls that can filter traffic
- Block unsolicited inbound traffic
- Filter link-local and multicast traffic

Implement secure address configuration

- Use stateful DHCPv6
- Enable RA Guard to secure SLAAC

Leverage IPsec for encryption

- Encrypt traffic

Only makes sense on site-to-site VPNs.

Control Neighbor Discovery Protocol (NDP)

- Deploy SEND (Secure Neighbor Discovery) to validate NDP messages
- Use NDP Inspection of NDP Guard to filter malicious solicitations/ advertisement message

Filter multicast and anycast traffic

- Restrict multicast traffic to authorized groups
- Carefully configure anycast addresses and monitor routing to avoid redirection

Disable unused features and tunnels

- Turn off IPv6 on devices when not used
- Disable legacy tunneling
- Monitor rogue tunnels

Monitor and log IPv6 traffic

- Employ intrusion detection
- Log IPv6 traffic
- Track global unicast addresses (2000::/3) for unexpected external connections

Secure routing protocols

- Use cryptography to authenticate routing protocols
- Filter router advertisements and routing updates from untrusted sources

Update firmware, software, and configurations

Segment networks and apply access controls

- Use VLANs or subnets to segment IPv6 traffic
- Apply access control lists (ACLs) to restrict traffic between segments

10 Implementing IPv6 in ABT Site

We are not fully rolling out IPv6 for the current release, but it is important to start acquiring knowledge and experience on the topic as soon as possible. To that end, we are deploying a toggle file for IPv6 functionality in ABT Site to allow engineers and commissioners in the field to work with IPv6. We are also offering a dual stack option in the field to ensure a smooth transition to IPv6.

BACnet/IPv6 and IPv4 are set to no by default. Devices ship with IPv4 functionality active and IPv6 functionality dormant unless explicitly enabled.

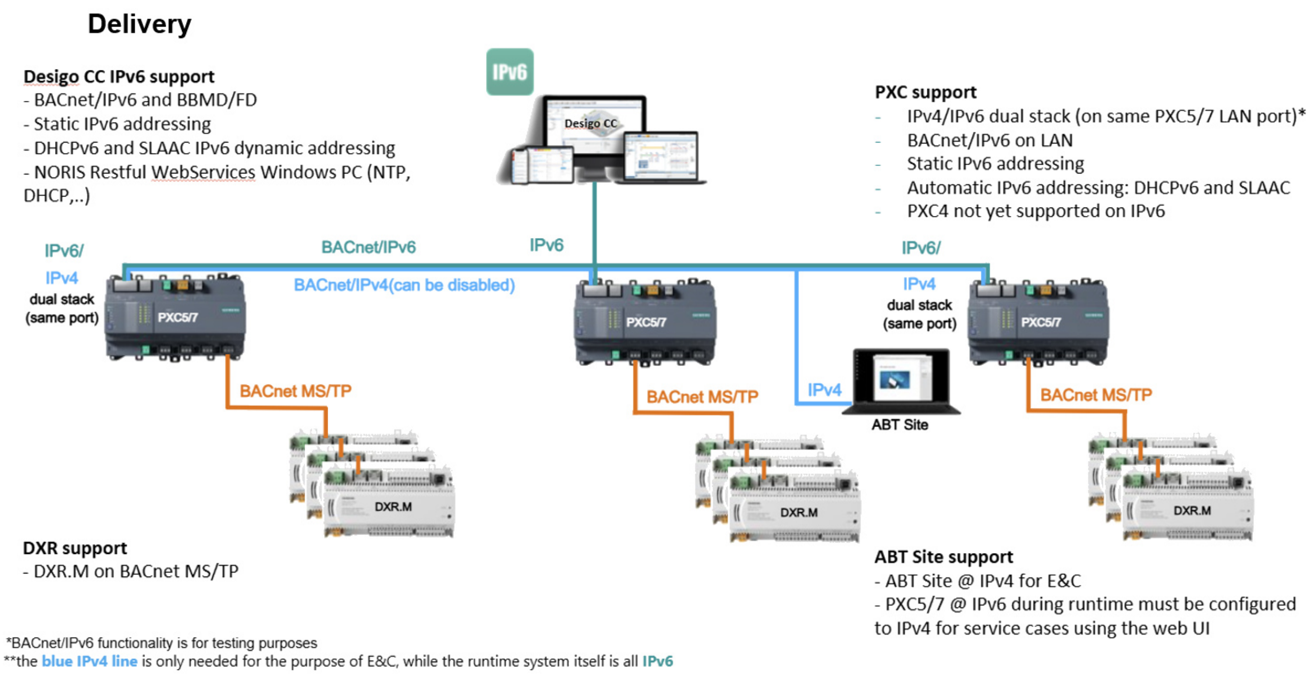
IPv6 support is available, but we recommend enabling BACnet/IPv6 functionality for testing only.

For ABT 6.1:

- BACnet/IPv6 configuration is not loaded by default.
- BACnet/IPv6 and IPv6 default set to **No**.
- Devices ship with IPv4 active and IPv6 functionality dormant unless explicitly enabled.

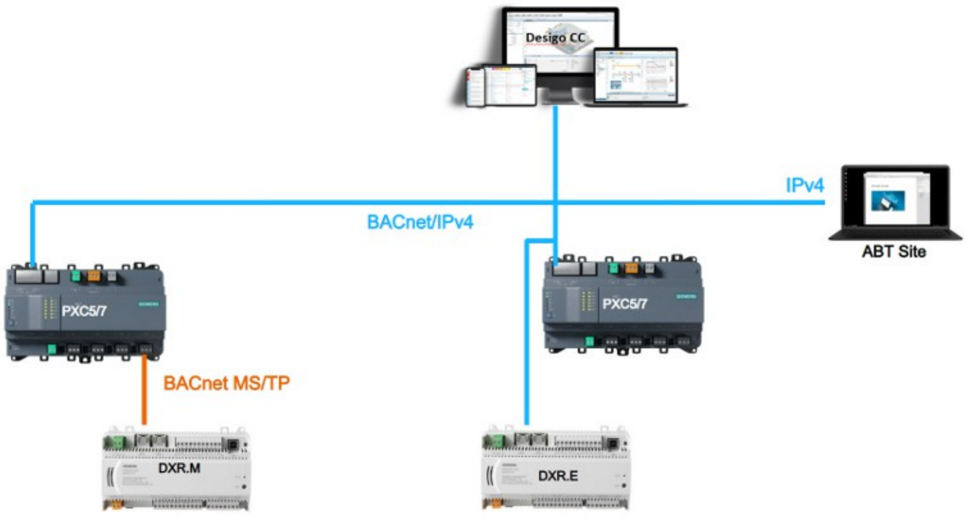
10.1 Topologies

Delivery



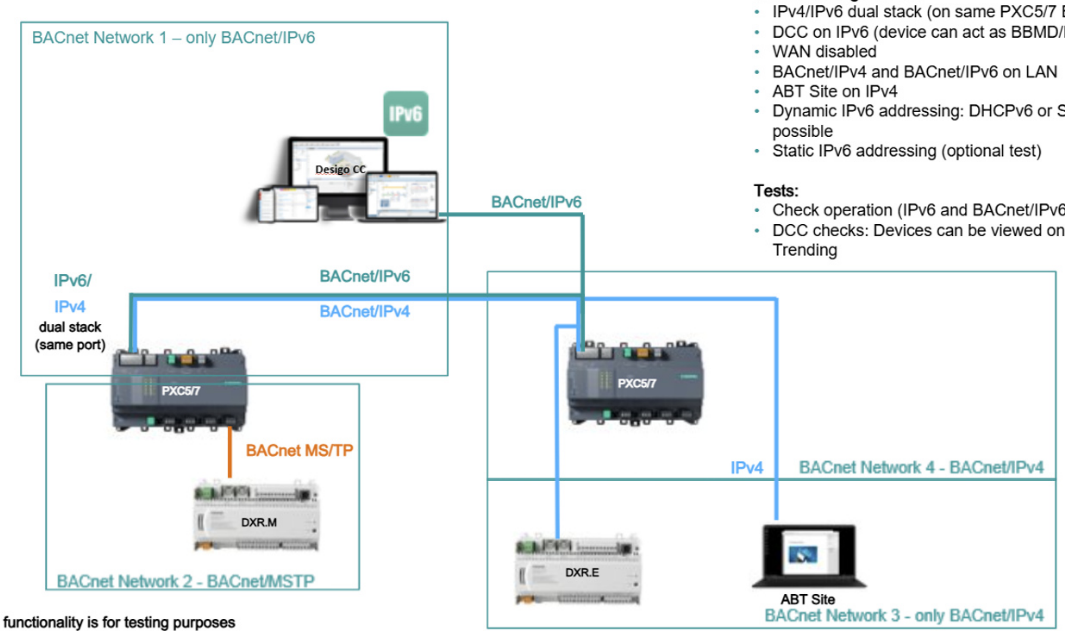
Start with IPv4 on LAN

Initial Configuration:
- BACnet/IPv4 on LAN
- DCC on IPv4
- WAN disabled
- ABT Site on IPv4



*BACnet/IPv6 functionality is for testing/experimental purposes

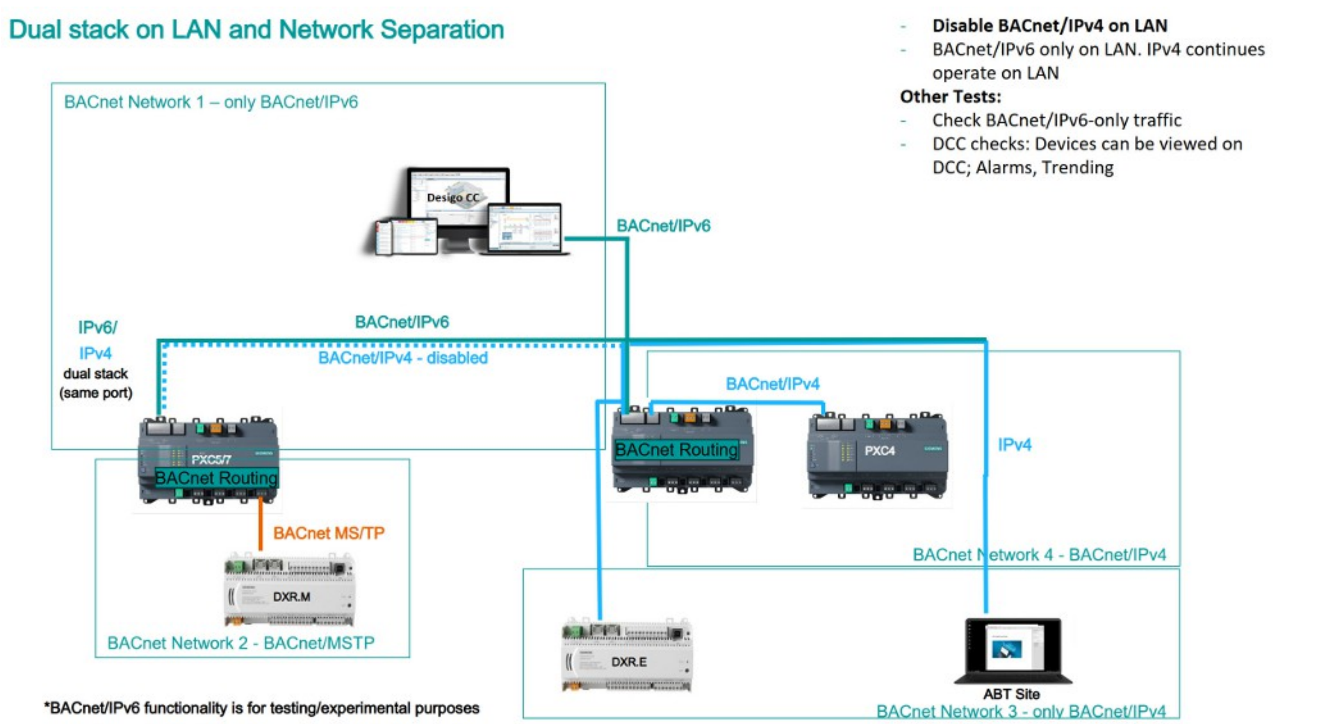
Enable Dual-stack IPv4/IPv6 on LAN

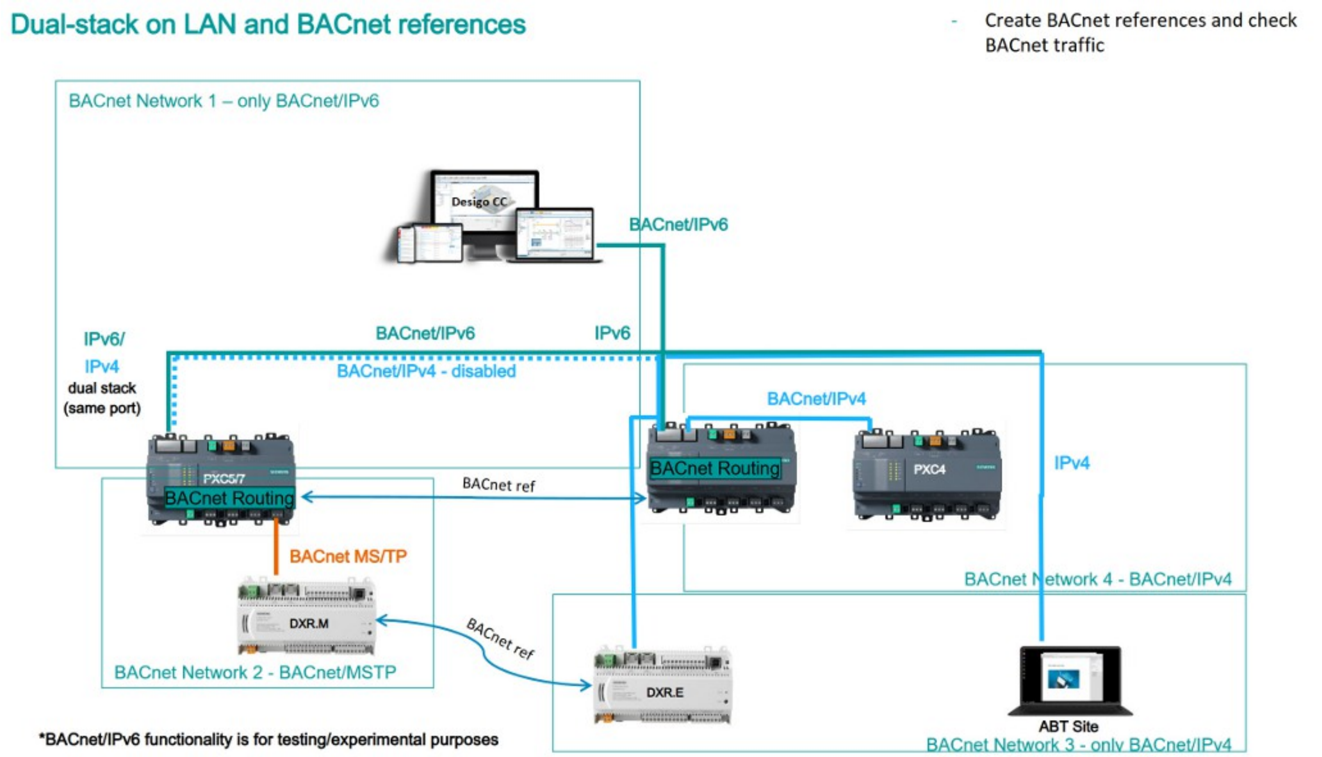


What to configure next:
- IPv4/IPv6 dual stack (on same PXC5/7 Eth. port)
- DCC on IPv6 (device can act as BBMD/FD)
- WAN disabled
- BACnet/IPv4 and BACnet/IPv6 on LAN
- ABT Site on IPv4
- Dynamic IPv6 addressing: DHCPv6 or SLAAC are possible
- Static IPv6 addressing (optional test)

Tests:
- Check operation (IPv6 and BACnet/IPv6 traffic)
- DCC checks: Devices can be viewed on DCC; Alarms, Trending

*BACnet/IPv6 functionality is for testing purposes





10.2 Supported address modes

Address type	Description
Link-local unicast	Device generates a network address that is only valid within the network segment or the host broadcast domain Not routable and starts with FE80 Typically used for demos, development, or testing and not recommended for end use.

Address type	Description
Static address	The user sets the IPv6 address, prefix length, and default gateway; generally the best option if SLAAC is not available.
DHCPv6	Requires a DHCP server The server assigns the device an IPv6 address, prefix length, and default gateway.
SLAAC	Automatically generates an IPv6 address, prefix length, and is assigned a default gateway based on information provided by its network. SLAAC is similar to DHCP on IPv4 in the sense that it can be used to generate global IPv6 addresses.

Issued by
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